

Zakaria RAHOU

Computer & Data Engineering Student — Polytech Annecy



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Profil :

Engineering student specializing in data engineering, software development, AI, and scientific analysis. Strong skills in data modeling, full-stack development, visualization, and building reliable, scalable systems.

ML & Data Science Interests :

Machine learning pipelines, data preprocessing & automation, feature engineering, visualization for complex structures, lightweight AI assistants, NLP & speech processing, graph analytics (NetworkX), reproducible modeling with Docker, ethical & transparent AI systems.

Skills :

- **Languages:** Python, Java, C++, SQL, JavaScript, TypeScript
- **Frameworks:** Next.js, React, Angular,
- **Build Tools & Front-end:** Vite
- **Deployment & Containers:** Docker, Vercel
- **Databases:** PostgreSQL, MongoDB
- **Tools:** Git, Excel

Projects :

- **Library Management Software Development :**
Analyzed municipal requirements for library management, including book, CD, and DVD loan tracking. Designed and implemented key functionalities to streamline operations.
- **Data Processing & Business Analysis - France Plomberie Chauffage (FPC) - Python (Pandas, OpenPyXL) • Excel**
Processed and cleaned customer, prospect, lead, and supplier datasets using Python (Pandas). Automated data transformation and exported structured results back to Excel using OpenPyXL. Produced clear summaries and KPIs to support operational decision-making.
- **Individual and Team Mini-Projects - Python • Java • C :**
Developed small-scale projects individually and collaboratively using multiple programming languages.
- **Team Project – Web Application for Academic Projects :**
Created a website to showcase academic projects (APP).
Managed site information, including participant lists, sessions, documents, a photo gallery, and interactive features. Ensured file structure compliance for site portability.
Optimized SQL queries to enhance database performance.
- **Team Interactive Digital Portfolio (Online CV) - Next.js • React • Tailwind**
Created a fully interactive digital portfolio.
Component-based architecture with optimized rendering.
Clean UX and responsive layout.
URL : <https://portfolio-beige-delta-89.vercel.app>
- **ThermoElite Corporate Website & Lead Generation System Next.js • React • Tailwind • Email API • Vercel**
Developed a commercial website for ThermoElite (heat-shrink confinement).
Built a responsive interface presenting services and use cases.
Added a lead-generation form sending client messages directly to company email.
Deployed on Vercel for high performance and reliability.
URL: <https://thermo-elite-website.vercel.app>
- **Dashboard for Student Competency Tracking - NetworkX • Plotly 3D • Python**
Modeled and analyzed graph structures using NetworkX.
Built interactive 3D visualizations (rotation, zoom, hover) with Plotly.
Styled the interface with **Shadcn UI** for a clean and modern user experience.

- **AI-Based Budget Management Assistant - Next.js, React, MongoDB & API Gemini**
Developed a smart assistant that analyzes user transactions.
Identified expense patterns and produced automated insights.
Deployed in a reproducible Docker environment.
- **Voice Assistant for Restaurants - Next.js • JavaScript • Gemini API • Twilio**
Created a voice interface for order management.
Integrated speech-to-text and natural language processing.
Designed conversational flows and real-time command handling.

Education:

Polytech Annecy | *Master of Science - MS, Computer Science and Data Engineering (2023 - Current)*

- Focused on data science, artificial intelligence, software development, and user-centric computing.
- Additional coursework in project management, digital transformation, and IT ethics.
- Practical experience in data processing, machine learning, and software engineering through hands-on projects and industry collaborations.

Université Grenoble Alpes | *Preparatory courses for Engineering Schools – Mathematics & Physics (2021 - 2023)*

- Intensive coursework in advanced mathematics, physics, and engineering sciences.
- Focused on problem-solving, analytical thinking, and scientific reasoning.
- Developed strong computational and modeling skills through rigorous training and projects.

Interests and Activities:

- **International Experience:** 1 year living in **Australia** + extensive travel, strong cross-cultural adaptability.
- **Sports Leadership:** Youth football coach — team management, mentoring, communication.
- **Hiking:** Monthly hikes in the Alps.